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Chapter 21: Financial Products and Practitioners—The Micro-Consequences of Fiber Inflation

Abstract

This chapter descends from the macro-level diagnosis of cognitive debt and the fiber framework to the micro level, exploring the structural consequences borne by financial products and practitioners amidst fiber inflation. First, this chapter dissects the de-anchoring pathways and regulatory calibration logics of three representative financial products (pooled funds, cryptocurrencies, and derivatives), revealing the pseudo-small fiber characteristic of pooled funds, the degenerate form of "fiber as the total set" for uncollateralized cryptocurrencies, and the tail risks inherent in excessively long derivative mapping chains. Second, rather than presupposing "several categories of practitioners," this chapter identifies, along the investment/mapping chain, the **key intermediary nodes where the actual degradation of "narrative load (ϕ) replacing meta-audit capacity (A)" occurs**, distinguishing two degradation pathways: Pathway ① Structural Compulsion (institutions/technology render a node *unable* to exercise meta-audit capacity) and Pathway ② Incentive Induction (a node *possesses the capacity but is induced by incentives to abandon* meta-audit capacity). Under this axis, bank wealth management relationship managers in the era of pooled funds are positioned as a historical sample specific to a particular period of Pathway ① (institutionally terminated after the end of the transition period of the new asset management regulations at the close of 2021), while sales/distribution nodes and asset management nodes present cross-national evidence demonstrating that the incentive degradation of Pathway ② remains widespread in the present. The

research indicates that when institutional calibration forces fiber to become visible, financial practitioners bear the brunt of the micro-costs of cognitive dissonance and cognitive debt liquidation—a substantial portion of the costs of systemic fiber inflation surfaces, from beyond the institutions' books, as the occupational circumstances of specific individuals.

Keywords

Fiber Inflation; Pooled Funds; Cryptocurrency; Cognitive Dissonance; Pseudo-Small Fiber; Meta-Audit Capacity

§1 Introduction: From Structure to the Individual

The preceding four chapters (17 to 20) diagnosed fiber inflation in financial markets from different facets: Chapter 17 established an overall diagnosis of financial markets within the fiber framework, Chapter 18 analyzed meta-audit at the institutional level (compensation and revolving doors), Chapter 19 diagnosed the spatialization of the mapping chain—speed limits, the quadruple audit dilemma, and the fundamental changes wrought by global continuous trading on the mapping chain structure, and Chapter 20 revealed how the ideological filter of the pricing system produces hidden subspaces.

These diagnoses all operate at the macro level—they analyze market structures, institutional designs, and epistemological constraints. But fiber inflation is not abstract: it is transmitted to investors through specific financial products and to practitioners through specific incentive mechanisms. The task of this chapter is to descend from the macro diagnosis to the micro level, answering two questions: First, what fiber states and regulatory responses do three representative financial products (pooled funds, cryptocurrencies, and derivatives) each exhibit? Second, how is fiber inflation transmitted from macro structures to specific practitioners—not through presupposed "categories of practitioners," but by identifying, along the investment/mapping chain, the key intermediary nodes where the $A \to \phi$ degradation actually occurs?

This chapter is divided into two parts. The first part (§2-§5) analyzes the de-anchoring pathways and calibration models of three financial products. The second part (§6), without presupposing categories of practitioners, identifies along the investment/mapping chain the **key intermediary nodes** where $A \to \phi$ degradation actually occurs—bank wealth management relationship managers in the pooled fund era (a historical sample of Pathway ①), sales/distribution nodes, and asset management nodes (current cross-national mechanisms of Pathway ②)—using the fiber framework to dissect the complete transmission chain of fiber inflation from the macro to the micro level. The common theme of both parts is: **The consequences of fiber inflation are ultimately borne by specific people—investors bear the pseudo-small fiber at the product level, and practitioners bear the cognitive dissonance at the structural level.** Within the overall framework of the transmission chain in the Financial Section, this chapter serves as the "Macro Structure → Micro Consequences" descent chapter: fiber inflation, produced at the macro level by incentive distortions, elongated mapping chains, institutional entropy increase, and ideological obfuscation, ultimately manifests as traceable micro-consequences in these three products and upon the practitioners at these key nodes.

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§2 Pooled Fund Products and the New Asset Management Regulations: The Calibration Logic of Look-Through Supervision

§2.1 The De-Anchoring Pathway of Pooled Funds

Pooled funds were once the most typical form of "de-anchored" product in China's financial system. Their operational mechanism was as follows: banks commingled multiple funds into a "pool" and invested the pool's capital in various underlying assets (non-standard debt, trust beneficiary rights, asset management plans, etc.). Maturity mismatch is the core feature of pooled funds—short-term funds (three-month wealth management products) matched with long-term assets (two-year non-standard debt), with liquidity maintained through "roll-over issuance."

Ji Min and Li Hongjin (2018), from the policy perspective of the People's Bank of China Research Bureau, pointed out that "under the aforementioned operational model of pooled funds matched with asset pools, banks did not actually bear or bore relatively little of these risks; simply renewing wealth management product issuance to maintain liquidity sufficed to mask asset-side risks" (author's translation)—meaning that the risk transfer within pooled funds was not an incidental management failure but an institutional logic embedded in its operational structure. What investors saw was a deterministic reading ("expected annualized yield of 4.5%"), but the underlying asset characteristics could involve high risk, maturity mismatch, and extremely poor liquidity.

Wei Lei, Wang Yitian, and Zhang Wei (2018) studied another form of fund-allocation arrangement—corporate group cash pools—showing that such arrangements, originally serving internal fund allocation within corporate groups, can likewise deviate from their real-economy functions, driven by the demand for "highly safe, highly liquid, and relatively high-yielding" (author's translation) quality financial assets, and evolve into shadow banking-style platforms for fund idling—a mechanism that is isomorphic, not a continuous evolution of the same institution. Zhang Yilin, Liang Wei, and Yu Yunjun (2023) further pointed out that the role of implicit guarantees in different developmental stages of China's shadow banking had structural differences: early-stage implicit-guarantee funds came from asset management institutions' own capital, serving as both a self-regulatory mechanism and a trust mechanism that compelled prudent investment; but when the pace of asset scale expansion outstripped the rate of own-capital accumulation, implicit guarantees degenerated into a risk-disguising "robbing Peter to pay Paul" scheme (author's translation)—it was precisely this turning point that provided the governance logic for the 2018 new asset management regulations' "mandatory breaking of implicit guarantees." Shao Xinjian, Dong Dingding, and Hong Junjie (2023), using data on government-affiliated financing platform products issued by trust companies for 178 city governments from 2010 to 2017, provided micro-level confirmation that investors' implicit-guarantee expectations and the buoyancy of urban land markets jointly constitute the two cornerstones of shadow banking debt pricing—the higher the urban land transaction price, the lower the credit spread charged to financing platforms, while failed land auctions significantly raise their financing costs.

Wen Xinxiang and Su Naifang (2018) characterized this overall structure as "Chinese-style shadow banking" or "the shadow of banks": unlike US shadow banking, which is dominated by non-bank institutions, China's shadow banking is led by commercial banks, where "channel business or interbank business between banks and non-bank financial institutions forms a financing chain, characterized by long processes and complex operations, not only circumventing regulatory requirements for traditional loans but also increasing financing costs" (author's translation)—"channel business disguisedly expands capital, causing funds to deviate from the real economy to the virtual to some extent" (author's translation). Gao Ran, Chen Chen, Zeng Hui, and Gong Liu-

tang (2018) revealed the macro driving force of this expansion using a dynamic stochastic general equilibrium (DSGE) model: the variation in the financing scale of China's shadow banking is **counter-cyclical**—when monetary policy tightens or loan-to-deposit ratio constraints become stricter, the dual credit constraints faced by commercial banks (capital adequacy ratio constraint and loan-to-deposit ratio constraint) actually squeeze more funds off-balance-sheet, causing shadow banking to accelerate its expansion during tightening periods. Si Dengkui, Li Yingjia, and Li Xiaolin (2022) added the perspective of banking competition: bank competition significantly curbs the shadow bankingization of non-financial enterprises by weakening the two mechanisms of credit distortion and regulatory arbitrage. In the language of the fiber framework, "channels" are the nested layers of the mapping chain—for each additional channel layer added, the mapping chain between the true state of the underlying asset (T) and the final reading (r) gains an intermediate node, causing the fiber to swell correspondingly.

In the language of the fiber framework (the static structural diagnostic tool established in Chapter 4): pooled fund products manufacture a pseudo-small fiber (δ_{pseudo}). The fiber of the reading seen by investors (expected yield, wealth management product net value) is extremely small ("this product should be worth this amount"), but the true fiber is enormous—because investors do not know the risk profile of the underlying asset portfolio, the extent of the maturity mismatch, or what would happen if roll-over issuance ceased. This evolution corresponds to the self-reinforcement of the pseudo-small fiber: the more successful implicit guarantees become → the more investors trust it → the more capital scale swells → the larger the true fiber becomes → the higher the institutional cost required to maintain the pseudo-small fiber → until it becomes unsustainable.

Chen Guojin, Jiang Xiaoyu, Liu Yanzhen, and Zhao Xiangqin (2021), based on bank micro-data, confirmed that "financial institutions with opaque assets excessively relying on wholesale funding for regulatory arbitrage" (author's translation) is the internal mechanism for systemic risk accumulation—banks with lower asset transparency engage in more aggressive regulatory arbitrage, contributing more to systemic risk. Pooled funds aggregate multi-layered assets and then redistribute them, rendering the true state of the underlying assets completely invisible, making auditing structurally impossible; their empirics provide panel data evidence for this structural diagnosis.

§2.2 Core Operations of the New Asset Management Regulations

In April 2018, the People's Bank of China, the China Banking and Insurance Regulatory Commission, the China Securities Regulatory Commission, and the State Administration of Foreign Exchange jointly issued the "Guiding Opinions on Regulating the Asset Management Business of Financial Institutions" (i.e., the "New Asset Management Regulations"), comprehensively regulating asset management businesses, including pooled funds—the joint issuance across departments was itself an institutionalized practice of financial regulatory coordination; on why coordination is necessary and how it can be proceduralized, Zhong Zhen (2018) offered a theoretical account from the perspective of regulatory power conflicts. The four core operational measures and their corresponding positions within the fiber framework are as follows:

- **Breaking Implicit Guarantees**: Forcing observable debt (D_c^{obs}) to align with true debt (D_c^{true}) (for the definition of cognitive debt (D_c), see Chapter 3, §2). Asset management products are prohibited from promising principal protection or guaranteed returns; investors must bear genuine risks. This is an institutional operation that forces observable readings to align with the true state.
- **Look-Through Supervision**: Eliminating the narrative barrier created by nested layers, directly exposing

underlying assets. Asset management products are prohibited from multi-layer nesting; underlying assets must be identified on a look-through basis. This is a physical truncation of the mapping chain—limiting the number of layers the mapping chain can stack.

- **Prohibiting Pooled Funds**: Disassembling the transmission channel into traceable point-to-point links. Financial institutions are prohibited from conducting or participating in pooled fund business characterized by roll-over issuance, collective operation, and separate pricing; each product must be separately managed, separately accounted, and separately booked. This is the dismantling of the self-referential structure of "calibrating this clock with other clocks"—when a pooled fund uses new investors' money to pay principal and interest to old investors, the product's "creditworthiness" comes from the readings of other funds within the pool (i.e., $\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$), rather than the true returns (T) of the underlying assets.
- **Net Value-Based Management**: Products are subject to net-value-based management, with net values promptly reflecting the returns and risks of the underlying assets; financial assets adhere to the principle of fair value measurement, with market-value measurement encouraged, and only closed-end products may use amortized cost measurement under the two prescribed conditions. The pseudo-small fiber is exposed amidst price fluctuations.

The epistemological mechanism of the pooled fund's "pseudo-small fiber" was pointed out early on in the self-reflections of industry quants. Derman and Wilmott (2009), in *The Financial Modelers' Manifesto*, described the isomorphic mechanism using a classic metaphor—"All models sweep dirt under the rug. A good model makes the absence of the dirt visible."—and directly criticized Collateralized Debt Obligation (CDO) pricing models for "sweeping under the model's rug all unknown dynamics; with the dirt ignored, all that's left is a single number, called the default correlation." Derman (2011), in *Models.Behaving.Badly*, further generalized this mechanism into the fundamental paradox of financial modeling: financial models are not replicas of physical laws but tools for approximate thinking, and their real danger lies not in the models themselves but in the idolatry of those who "confuse the model with the world." The "expected yield of 4.5%" of pooled funds is the Chinese version of this idolatry—a number endowed by the institutional structure with epistemological authority exceeding what it can bear.[1]

The transition period of the new asset management regulations was originally set to end at the close of 2020 and was extended to the end of 2021 due to the pandemic; after the transition period, the regulations entered a phase of normalized enforcement. Subsequently, regulation continued to intensify. In October 2025, the National Financial Regulatory Administration (NFRA) issued the "Administrative Measures for Asset Management Trusts (Draft for Comments)," further clarifying that trust companies shall not conduct pooled fund business nor set expected yields. In December 2025, the NFRA issued the "Administrative Measures for Information Disclosure of Asset Management Products of Banking and Insurance Institutions," unifying the information disclosure standards for three types of products, effective from September 1, 2026.

[1]The Shareholder Value Maximization (SVO)-Efficient Market Hypothesis (EMH)-Performativity triad diagnosed in Chapter 20, with its L2-level preference for being "quantifiable on a quarterly basis," provides the legitimizing framework for the "expected yield" narrative of pooled funds—the macro-level SVO filter stipulates that "only dimensions quantifiable on a quarterly basis enter the visible subspace," the meso-level pooled fund structure exhibits an isomorphic information-compression mechanism, and the micro-level relationship manager executes the daily operation of narrative load (ϕ) replacing meta-audit capacity (A) at the breakpoint of this mapping chain (see §6). The attribution of "bank endorsement" directly to the operation of narrative load (ϕ) in this section is consistent with the diagnostic hierarchy of Chapter 20—the SVO-EMH-Performativity triad (the meta-rule at the ϕ'' level) provides a legitimizing framework for the circulation of first-order narrative load (ϕ') (see Chapter 20, §3.2 and §5.1 for details).

§2.3 Empirical Effects and Calibration Costs

Academic research has confirmed the calibration effects of the new asset management regulations. Li Jinsheng (2025), based on panel data from 133 commercial banks, found that the new regulations effectively reduced banks' risk-taking levels. Zhao Dengpan and Wang Xiaofang (2025) further confirmed that the regulations reduced bank risk-taking through the leverage ratio channel and prompted banks to increase the proportion of corporate loans.

Fang Yi and Chen Min (2019) provide the more general mechanism background of risk-taking: bank risk-taking measured by the market-based asset variation rate is forward-looking, verifying that Minsky's Financial Instability Hypothesis holds for China—the procyclical phenomenon whereby low volatility during economic expansion induces excessive bank risk-taking and intensifies systemic risk accumulation, with China's financial cycle exhibiting a medium-term frequency of roughly eleven years. Prior to the new regulations, Xiang Houjun and Yan Yu (2017), based on panel data from 100 commercial banks spanning 2005–2015, had already pointed out the dual effect of wealth management product development on bank risk-taking: the benefit-seeking effect of wealth management products reduced bank risk-taking, but the advancement of interest rate liberalization, by promoting competition, increased risk-taking—both effects amplify the impact of the monetary policy stance on bank risk-taking. This finding provides the antecedent context for understanding the calibration logic of the new regulations: the new regulations are essentially an institutional intervention to sever the transmission of risk from off-balance-sheet to on-balance-sheet, implemented after the scale of wealth management products exceeded the banks' own capacity to absorb risk.

Tang Sheng and Rao Pingui (2025) found that after the implementation of the new regulations, listed companies that had previously participated in larger-scale shadow banking activities significantly increased their supply of trade credit—a shift consistent with the reallocation of funds from shadow banking business to the real-economy operational chain. Cai Guowei, Zhu Yadian, and Zhou Tong (2022), based on a quasi-natural experiment of the investment income tax reform, provide corroborating evidence from the fund allocation side: the liquidity supplied by the wealth management product market enabled enterprises under stronger financing constraints to obtain more short-term debt financing and lowered their short-term financing costs—under conditions of strictly controlling underlying product risk and fundraising use, the wealth management product market generated positive spillovers for the real economy.

But calibration has costs. Zhou Shangyao and Wang Sheng (2021), using a DSGE model incorporating a shadow banking run mechanism to measure systemic financial risk levels, found that, since 2007, the systemic risk of China's shadow banking sector "peaked in 2011 and 2013, corresponding to the two 'money shortage' events respectively" (author's translation); counterfactual policy simulations indicate that "every 10% reduction in systemic financial risk through regulatory policy leads to a long-term output reduction of approximately 0.18%" (author's translation)—in the model's counterfactual simulation, calibration trades output losses for risk reduction.

Xiang Houjun and Zhou Xiong (2022), from the perspective of liquidity hoarding, revealed another pathway cost of calibration: research based on data from 166 commercial banks found that "financial regulation of shadow banking weakens the negative impact of shadow banking on banks' liquidity hoarding levels, causing banks' liquidity hoarding levels to rise, thereby affecting the flow into the real economy" (author's translation)—after regulation cut off shadow banking channels, funds did not automatically flow to the real economy but partially

remained lodged within the banking system. Liu Chong, Pang Yuanchen, and Liu Liya (2022), based on bond market transaction data, found that after the new asset management regulations were issued, the transmission efficiency from short-term money market rates to bond rates declined significantly, while structural monetary policy tools such as Medium-term Lending Facility (MLF) collateral expansion played a countervailing, restorative role for transmission efficiency—the friction of calibration likewise appears in price-based transmission channels.

Bu Zhenxing (2021), using the extension of the new asset management regulations as a case study, documented the transitional resistance: difficulties in disposing of existing assets, low acceptance of newly issued products, and insufficient institutional motivation for transformation, which led to the extension of the transition period from the originally planned end of 2020 to the end of 2021. The quantitative estimate by Lu, Zhao & Niu (2025) is the most direct: after the implementation of the new asset management regulations, the bond credit spread between Chinese private enterprises and state-owned enterprises widened by an average of 54 basis points—while regulation reduced shadow banking risks, it pushed up private enterprises' financing costs relative to SOEs.

In the fiber framework, these costs are not "side effects" of calibration but rather the intrinsic cost structure of calibration itself: calibrating the clock requires severing the de-anchored structures; the severing process itself generates friction (output loss, liquidity hoarding, transitional pains, rising financing costs), and the costs of this friction are disproportionately borne by specific groups (private enterprises, small and medium-sized banks, practitioners during the transition period).

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§3 Cryptocurrencies: Regulatory Choices When Fiber is the Total Set

(The institutional logic of China's ban on cryptocurrencies can be positioned as "direct truncation"—direct intervention at the institutional level when neither the compensation loop nor crash calibration is available for such assets. This chapter deepens that diagnosis into the epistemological basis that "when fiber is the total set, the mapping function does not exist.")

§3.1 The Epistemological Status of Cryptocurrencies

Cryptocurrencies hold a unique diagnostic position within the fiber framework. For virtual currencies like Bitcoin and Ether, what is the true state (T) of the underlying asset? If the true state (T) does not exist—no cash flow from the real economy, no genuine corporate profits, no repayment capacity of underlying loans—then what is the input to the mapping function (π)?

The answer is: Cryptocurrencies are "mappings without an underlying asset." In the language of fiber, this is the **extreme case where fiber is the total set**—a reading (r, the price of Bitcoin) is compatible with all possible true states because there is no true state to anchor it. Here "fiber is the total set" is an epistemological intuition: the reading excludes no hypothesis about underlying value, so every hypothesis remains compatible with the reading; it is not the strict value of the inverse image (which is undefined when the mapping function does not exist), but the degenerate form of the fiber concept for this object. (For some algorithmic stablecoins, the mapping function exists but its inputs are entirely self-referential—the pricing rule itself constitutes a self-referential recursion containing no anchor external to the system; their fiber genuinely exists yet is equally

unable to trace back to a truth external to the system, and this chapter treats both together.) The price of a cryptocurrency is not "discovered" but "formed"—through the collective beliefs and self-fulfilling expectations of market participants.

Soros (1987/2003), in *The Alchemy of Finance*, proposed the concept of "reflexivity," which provides a practitioner's theoretical foundation for the diagnosis that "prices are formed": financial market participants influence the market through their expectations, and the market, in turn, shapes expectations; this self-reinforcing feedback loop does not converge to a stable equilibrium—in Soros's own words, markets "exhibit no tendency towards equilibrium. Indeed, it makes more sense to claim that markets tend towards excesses, which sooner or later become unsustainable"; its root lies in the fact that "thinking is part of the reality people have to think about." For assets with underlying cash flows, reflexivity is merely a deviation around the true state; for uncollateralized cryptocurrencies, reflexivity becomes the **dominant** driver of price—reflexivity without a true state as an anchor is the extreme case where fiber is the total set. Soros's reflexivity theory and MacKenzie's (2006) performativity theory, discussed in Chapter 20, §4.1, mirror each other: the former describes "expectations shaping the market" from a practitioner's perspective, while the latter describes "models shaping the market" from a sociological perspective; both point to the same structure—the readings of the financial clock possess ontological productivity, rather than being mere observations.

This diagnosis requires a crucial precise distinction. For uncollateralized cryptocurrencies (Bitcoin, Ether, etc.), the mapping function does not exist—the reading (price) is compatible with all possible true states because no true state can anchor the reading. Fiat-collateralized stablecoins (USDT, USDC, etc.) are different: they hold US dollars or dollar-denominated assets as reserves, and their mapping chain is—stablecoin price → USD reserves → Federal Reserve balance sheet → US government's taxing power. The ultimate anchor of this mapping chain is sovereign power, not the blockchain. Stablecoins are a digital extension of the dollar ecosystem, not a triumph of "decentralization." Their diagnostic position within the fiber framework is entirely distinct from uncollateralized cryptocurrencies: for the former, the mapping function exists (fiber is non-singleton, but the anchor is traceable); for the latter, the mapping function does not exist (fiber is the total set). Therefore, what this chapter refers to as "cryptocurrencies as noise where fiber is the total set" specifically denotes uncollateralized virtual currencies like Bitcoin and Ether; fiat-collateralized stablecoins like USDT and USDC, with their mapping chains anchored in sovereign power, belong to the former, anchor-traceable category. This ontological difference ("mapping function does not exist" vs. "mapping function exists but fiber is non-singleton") is sufficiently significant to serve as a key distinguishing point in this chapter from §2 (pseudo-small fiber of pooled funds) and §4 (derivatives where fiber swells with nesting layers but the underlying true state (T) exists).

As of December 2025, the total assets of dollar stablecoins exceeded 270 billion US dollars, with reserve positions in US short-term Treasury bills of 153 billion US dollars, surpassing the holdings of some major foreign investors and domestic money market funds (Ahmed & Aldasoro, 2025). This data directly supports the diagnosis that "fiat-collateralized stablecoins establish a mapping relationship with the true state via USD reserves."

§3.2 China's Path: Comprehensive Prohibition and Its Epistemological Basis

China's regulatory path is one of "comprehensive prohibition." In February 2026, eight departments, including the People's Bank of China, jointly issued the "Notice on Further Preventing and Addressing Risks Related to Virtual Currencies and Related Matters" (Yinfa [2026] No. 42), reiterating that virtual currencies do not have the same legal status as fiat currency; absent the consent of the relevant departments in accordance with

laws and regulations, no domestic or foreign entity or individual may issue RMB-pegged stablecoins overseas; for the first time, it clarified the regulatory framework for Real World Asset (RWA) tokenization, stipulating that RWA activities conducted within Chinese territory are prohibited in principle, with the exception of related business activities conducted through specific financial infrastructure with the consent of the competent business authorities in accordance with laws and regulations.

In fiber language, Chinese regulators have determined cryptocurrencies to be "noise where fiber is the total set" and thus exclude them from the financial system. This is not "rejecting innovation"—interpreted through this chapter's epistemological framework, Chinese regulation has set strict financial-system access boundaries for both unanchored readings and asset-tokenization activities; "whether a reading can be traced back to an independent anchor within a finite number of steps" is this chapter's criterion for analyzing these boundaries, not a self-stated rationale in the regulatory documents. A financial clock must be traceable back to the sun (the true state) within a finite number of steps—a clock without a sun is not a clock; it is noise.[2]

§3.3 The Institutional Oscillation in the United States

In 2025, US crypto-asset regulation underwent a dramatic directional shift. Before this, during Gary Gensler's tenure, the US Securities and Exchange Commission (SEC) brought a large number of enforcement actions against crypto firms. Gensler resigned in January 2025, and Paul Atkins assumed the role of SEC Chair in April, shifting the direction from regulation by enforcement to empowering compliance (SEC, 2025). On July 18, 2025, the GENIUS Stablecoin Act (U.S. Congress, 2025) was signed into law, establishing reserve and regulatory requirements for payment stablecoins; its main provisions take effect on the earlier of 18 months after enactment or 120 days after the issuance of final implementing regulations.

In October 2024, the crypto-native platform Crypto.com acquired the SEC-registered broker-dealer Watchdog Capital and officially launched stock trading in January 2025—becoming a representative case of a crypto-native platform extending into regulated securities business and an epitome of mixed operations extending into the crypto space. A Bank for International Settlements (BIS) working paper, based on an empirical analysis of cross-border Bitcoin, Ether, and stablecoin flows, found that capital flow management measures "appear ineffective" in curbing crypto-asset trading, with trading volumes for certain crypto assets even increasing following the imposition of controls (Auer, Lewrick & Paulick, 2025).

The core difference between China and the US lies in this: Chinese regulation is epistemologically grounded, drawing its boundaries based on the definition of "what a financial product is," resulting in a long-term stable stance (unchanged from 2017 to the present); US regulation is based on the political cycle, with its direction oscillating significantly between "enforcement—relaxation—re-enforcement—re-relaxation." In fiber language: China's regulation is "prohibiting noise from entering the room," while US regulation is "allowing noise to enter, but potentially changing the rules at any time."

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[2] "Epistemological stance," in the fiber framework, refers to the regulator's fundamental judgment concerning "what conditions a financial clock must satisfy to be accepted as a legitimate reading"—China's stance is that "the reading must be traceable, within a finite number of steps, to an anchor independent of the financial system" (finite traceability renders it legitimate; untraceability renders it noise); the US stance oscillates in the domain of crypto-assets, altering the boundary of "what counts as a financial product" with the political cycle (see §3.3). The distinction between this concept and the "ideological apparatus" diagnosed in Chapter 20 is: the former is the criterion by which the regulatory subject judges the legitimacy of a mapping function, while the latter is the naturalized presupposition of the mapping function itself.

§4 Derivatives and Asset Securitization: When Clocks Are Permitted to Exist

Derivatives and asset securitization are "legitimate" clocks within the financial system—they have underlying assets (equities, debt claims, loans, commodities), but fiber inflates with successive layering. Unlike cryptocurrencies, the underlying true state (T) for these products exists; therefore, China's regulatory choice is not "comprehensive prohibition" but "prudent development."

In October 2025, the recommendations for the "15th Five-Year Plan" incorporated derivatives and asset securitization into the national strategic deployment—"steadily developing futures, derivatives, and asset securitization" (Central Committee of the Communist Party of China, 2025; author's translation). The outstanding stock of China's interbank bond market reached 173.7 trillion yuan at the end of 2025, accounting for 88.3% of the national bond market. Interbank market derivatives trading volume reached 258.5 trillion yuan in 2025 (National Association of Financial Market Institutional Investors, 2026). The asset securitization market also maintained growth; in 2025, the issuance scale of asset-backed securities (ABS) reached 2.28 trillion yuan (China Financial Information Network, 2026); publicly offered Real Estate Investment Trusts (REITs) have cumulatively raised over 200 billion yuan since the pilot program began in 2021 (Guotai Haitong Asset Management, 2026), becoming a new type of mapping channel connecting the true state of real estate with financial market readings.

China's prudent development has clear institutional bottom lines: limiting the number of nested layers (look-through supervision written into law), limiting leverage ratios (macro-prudential assessment), and strengthening information disclosure (net-value-based management, related party transaction disclosure). The core logic is: clocks can be built, but they must be traceable back to the sun within a finite number of steps. Pan Gongsheng (2025) pointed out that China is one of the few economies to have fully implemented Basel III. For example, the normalized issuance of publicly offered REITs in 2026 requires that the cash flows of the underlying assets (infrastructure projects) be independently accountable and traceably auditable—this is the institutionalization of the "finite-step traceability" principle at the product level. However, the current global financial stability framework faces a tendency towards a "race to the bottom"—the enforcement of international regulatory standards like Basel III is subject to oscillations influenced by domestic political factors; and whether regulatory standards can be effectively implemented also depends on the incentive-compatible structure of the regulatory system itself (Xu Zhong, 2018).

A comparison with international markets is equally important. In 2025, the new issuance of US Collateralized Loan Obligations (CLOs) reached approximately $201.5 billion, surpassing the 2024 record (Bloomberg News, 2025b; data as of mid-December 2025); alternative asset managers such as Blackstone, Apollo, Ares, and Blue Owl sold private credit CLOs at the fastest pace in history—approximately $30 billion had been sold as of September 2025 (Bloomberg News, 2025a). This constitutes a complete chain from underlying opacity to rating-based reading: the newly added private credit subset within the CLO market (having accounted for approximately 18% of the US private credit market in 2024) is growing particularly rapidly, its underlying assets having lower transparency compared to traditional Broadly Syndicated Loan (BSL) CLOs, and this portion of opaque products, read with AAA ratings, precisely meets global capital's thirst for "certain high yields."

Numerical analyses by Kim (2024) published in the *Journal of Futures Markets* show that CLO tranche valuation models that abandon the Large Homogeneous Portfolio (LHP) assumption yield higher tranche spreads, with the difference especially pronounced for lower-rated tranches; the results further suggest potential mispricing of spreads when valuing higher-rated tranches with LHP models. In the language of this book's fiber

framework, this provides an analytical instance of L2-level pseudo-calibration at work in the international market: the assumption structure of the valuation model may push the spread risk it excludes into a hidden subspace. Compared with China's prudent development path, the explosive growth of the US CLO market demonstrates the consequences of an alternative institutional choice: the mapping chain lacks institutionally imposed limits on its length, and pseudo-calibration is driven by market demand rather than constrained by institutions.

Bookstaber (2007), in *A Demon of Our Own Design*, offered a structurally isomorphic warning from an insider's perspective (former risk management head at Morgan Stanley, hedge fund manager): the systemic risk of the financial system **stems primarily from financial innovation itself, rather than from fluctuations in economic fundamentals**—the turmoil in financial markets stems more from the financial instruments (derivatives) originally designed to control risk than from economic fundamentals that have become more stable. He wrote:

> One of the curious aspects of worsening market crises and financial instability is that these events do not mirror the underlying real economy. In fact, while risk has increased for the capital markets, the real economy, the one we live in, has experienced the opposite. In recent decades the world has progressively become a less risky place, at least when it comes to economics. In the United States, the variability in gross domestic product (GDP) has dropped steadily. Year by year, GDP varies half as much as it did 50 years ago. (Bookstaber, 2007)

He attributed this to the "complexity × tight coupling" of the derivatives chain—once a single point fails, the entire chain, due to tight coupling, cannot be locally isolated. Published in 2007, the book issued a structural warning ahead of the 2008 crisis. His diagnosis provides the theoretical basis for China's calibration prescription of "limiting chain length": the longer the mapping chain, the more compliant each single layer may appear, but the tail risk of the entire chain accumulates at a pace far beyond what any single layer's compliance checks can cover.

It should be noted that Bookstaber's "innovation itself is the source of risk" is a strong-form judgment, not a scholarly consensus. The Chicago School's Miller (1995) took the opposite position: the rapid spread of the three classes of derivatives (exchange-traded futures and options, swaps, and structured notes) occurred precisely because they "satisfied an important business need," enabling firms and banks to manage at low cost the risks that had long troubled them; the massive losses at Procter & Gamble, Metallgesellschaft, and the like were individual misjudgments and did not constitute a systemic "derivatives-induced collapse." He argued further that "crashes in financial markets are not exogenous calamities like earthquakes. They are policy disasters"—stemming mostly from policy errors such as excessive central-bank tightening, rather than from private transactions themselves. This book presents the two positions symmetrically: Bookstaber's "complexity × tight coupling" diagnosis and Miller's "genuine need + policy attribution" defense each have their empirical support; the fiber framework takes from the former the structural judgment that "an over-lengthened mapping chain lets tail risk escape the institutional field of view," but does not deny the latter's functional judgment that "derivatives themselves meet genuine risk-management needs"—the true disagreement between the two lies not in whether derivatives are useful, but in whether an over-lengthened mapping chain turns "useful" into "unauditable."

In the fiber framework, the uniqueness of this "third state" (derivatives) lies in this: what it confronts is not a "black box"-style structural invisibility, but a legitimate yet excessively long mapping chain—the underlying

true state exists and is traceable, but the number of nested layers exceeds the audit capacity of any individual.

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§5 Comparison and Transition of the Three Calibration Models

Three financial products correspond to three different calibration models:

Pooled Fund Products (Look-Through + Prohibition + Net-Value-Based Management): Regulation makes fiber visible by breaking implicit guarantees, limits mapping chain length through look-through supervision, and severs the self-referential loop by prohibiting pooled funds. Calibration results have been validated through academic research, but there are costs—rising relative financing costs for private enterprises.[3] The core characteristic: calibration originates from proactive institutional intervention, not crisis compulsion.

Cryptocurrencies (Prohibiting Noise vs. Institutional Oscillation): China deems fiber to be the total set and excludes it from the financial system; the US lacks a consistent framework due to political cycles. The core characteristic: calibration exposes a deep divergence in financial regulation—what counts as a "financial product."

Derivatives (Permitting Clock-Making, Limiting Chain Length): Neither laissez-faire nor comprehensive prohibition, but rather accepting the utility of financial clocks while mandating an upper limit on traceability. The core characteristic: the "third state."

More importantly, calibration requires continuous work. From the introduction of the new asset management regulations in 2018 to the writing of look-through supervision into law in 2026 (Standing Committee of the National People's Congress, 2026), each tightening measure built upon the previous one, never exhibiting a "tighten then relax" pattern. This corroborates the core thesis of Chapter 17: calibration is anti-entropic work that must be continuously sustained.

Minsky's (1982/2016) "Financial Instability Hypothesis," presented in his collected volume *Can "It" Happen Again?*, provides a dual theoretical basis for "calibration requires continuous work": the endogenous dynamics of the financial system tend to evolve from hedge finance → speculative finance → Ponzi finance; **stability itself breeds instability** ("stability is destabilizing"—the precise origin of this formulation is Minsky's 1986 monograph *Stabilizing an Unstable Economy*)—once regulation slackens, market participants will interpret the new stable state as "risk has disappeared," subsequently adopt more aggressive leverage strategies, ultimately pushing the system back to the brink of instability. This is structurally isomorphic with the "calibration-entropy" binary opposition of this chapter: calibration (regulatory work) and de-anchoring (financial innovation's endogenous propensity toward complexity) constitute a perpetual, unceasing dynamic game; any narrative of a "one-time solution" contradicts the fundamental dynamics of the financial system.

The fiber diagnosis of the three products is now complete. But products do not operate themselves—they are sold, bought, and managed by people. The next section shifts from products to people: without presupposing "several categories of practitioners," it identifies, along the investment/mapping chain, the key intermediary

[3]The widening of credit spreads is simultaneously influenced by multiple factors, including the advancement of interest rate liberalization, the monetary policy cycle, and global capital flows. The 54 basis point estimate by Lu et al. is a difference-in-differences estimate controlling for other factors and cannot be directly equated with "private enterprise financing costs have thus risen"; a clear attribution of the calibration costs of the new asset management regulations requires distinguishing the direct effects of institutional calibration from the indirect effects of parallel structural changes.

nodes where the $A \to \phi$ degradation actually occurs, examining how the macro diagnosis of fiber inflation produces observable micro-consequences at specific nodes.

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§6 Key Intermediary Nodes: $A \to \phi$ Degradation Along the Investment Chain

The diagnosis of the three products revealed how fiber is manufactured at the product level, but products are operated by people. This section does not presuppose "several categories of practitioners"; it first establishes an abstract axis and then fills in specific nodes as instances.

Axis: Along the mapping/investment chain, each intermediary node faces the choice between "exercising meta-audit capacity (A)" or "producing narrative load (ϕ)"; the node's structural position and incentives determine whether it degrades. The inflation of fiber, $\pi^{-1}(r)$, is the cumulative result of the $A \to \phi$ degradation along the chain. Which nodes undergo degradation is determined by chain topology and incentive structures, not by a priori stipulation. Ma, Tang, and Wang (2024), in their review of mutual fund incentives, point out that investors do not directly face the final decision-maker but experience a double-layered delegation: investors → fund advisors → portfolio managers—this provides an organizational methodology basis for "finding key degradation nodes along the chain."

Degradation is divided into two pathways (consistent with §7 and the dichotomy in Chapter 18, but **not tied to historical periods**):

- **Pathway ① Structural Compulsion**: Institutions/technology render a node *unable* to exercise meta-audit capacity (A)—the mapping chain is broken, and the underlying is untraceable. Bank wealth management relationship managers in the pooled fund era are its historical sample (§6.1).
- **Pathway ② Incentive Induction**: A node *possesses the capacity but is induced by incentive structures to abandon* meta-audit capacity (A). Sales/distribution nodes (§6.2) and asset management nodes (§6.3) are its current, cross-nationally prevalent instances; their empirical demonstration is found in Chapter 18.

Following this axis, the discussion below first takes the pooled fund front office as a historical sample of Pathway ①, then examines sales nodes and asset management nodes as current cross-national mechanisms of Pathway ②, and finally returns to the current situation of Chinese front-office practitioners after the net-value transformation.

§6.1 Historical Sample of Pathway ①: The Mapping Chain Structure in the Pooled Fund Era (Pre-New Asset Management Regulations, Terminated by Net-Value Transformation)

The pooled fund front office diagnosed in this subsection is a **product of a historically specific period prior to the 2018 new asset management regulations**—the net-value transformation has institutionally terminated the specific structure of "fund commingling + implicit guarantees." It is placed here not because it represents the general state of the current front office, but because it is a **clean historical illustration of Pathway ① "Structural Compulsion"**: the mapping chain was technically severed, and even if practitioners had the will, they could not trace the underlying. The predicament of the current front office is a different matter (Pathway ②, see §6.2 and §6.4) and cannot be conflated with this historically and institutionally terminated phenomenon.

Pooled fund relationship managers faced a predicament of "mapping chain breakage": the underlying was structurally invisible. This is the clean form of Pathway ① "Structural Compulsion": not unwilling to audit, but

structurally unable to audit.

The pooled fund product was not merely a form of financial product but also a specific mode of labor organization. Zhang Jiaming (2022) provides the background structure for the micro-diagnosis here from a macro-transmission perspective: his research finds that "the bank risk-taking channel of monetary policy manifests as risk-taking transmitting from the liability side to the asset side of banks" (author's translation), and that "both asset-side and liability-side bank risk-taking are transmitted to shadow banking through the capital constraint effect" (author's translation)—shadow banking was not an isolated phenomenon but an institutional byproduct of commercial banks shifting risk off-balance-sheet under regulatory pressure. Relationship managers were the micro-terminals of this macro-transmission chain. Before the new asset management regulations, the daily work of a bank wealth management relationship manager was embedded in the mapping chain structure of pooled funds. The typical scenario he faced was: selling a wealth management product with an "expected annualized yield of 4.5%" to a client. The client asks: "Is this product safe?" The relationship manager's standard answer is: "This is our bank's product, it's very safe."

In the fiber framework, this scenario contains a complete micro-transmission chain.

Level L0 (Mapping Chain Structure): The relationship manager does not know what the underlying assets are. Pooled funds mix multiple funds and invest them in underlying assets like non-standard debt and trust beneficiary rights, with maturity mismatch maintained by roll-over issuance. The relationship manager faces a "black box"—the mapping chain between the reading he sees (expected yield of 4.5%) and the true state of the underlying assets (T) has been broken. It is not that he "chooses not to look," but that he is **structurally unable to see**. The asset commingling mechanism of pooled funds makes it technically impossible for practitioners to trace the underlying, even if they have the will to audit—this is a different mechanism from the fund managers diagnosed in Chapter 18 who "possess audit capacity but are incentivized to forgo auditing." The former is structural invisibility; the latter is incentive-based non-auditing.

Level L1 (Optimization Dynamics): The relationship manager's incentive structure is linked to sales volume, not to the quality of the underlying assets. His Key Performance Indicator (KPI) is "how much was sold," not "was the right product sold." In the language of the Macro-Cognitive Accounting Framework (MCAF), his governance clock (τ_g) is captured by the sales target—the decision frequency is determined by the sales cycle, not by audit needs.

Level L2 (Pseudo-Calibration): The relationship manager substitutes underlying asset analysis with "bank endorsement." When he says, "This is a bank product," he is effectively replacing audit capacity concerning the true state, i.e., meta-audit capacity (A), with the narrative load (ϕ) of institutional reputation. This is not intentional deception—he genuinely believes the bank will bail it out because "implicit guarantees" were an implicit contract operating in the industry for decades. The replacement of meta-audit capacity by narrative load is not a moral failure but a structural consequence of institutional design.

The pooled fund front office is a clean sample of Pathway ①. However, having been terminated by the net-value transformation, it can no longer bear the burden of arguing for "the structural predicament of current practitioners." The next two subsections turn to Pathway ②: even outside "black box" product structures like pooled funds, sales nodes and asset management nodes still undergo $A \to \phi$ degradation due to incentive structures—these nodes *possess the capacity* to audit the underlying, yet are induced by commissions, suitability deficiencies, the absence of fiduciary duty, assets-under-management (AUM) incentives, and ranking pressures

to forgo auditing. The cross-national evidence for both demonstrates that the practitioners' predicament is not unique to the specific product structure of pooled funds, nor is it unique to China.

§6.2 Current Cross-National Mechanism of Pathway ② (I): Sales/Distribution Nodes

Sales/distribution nodes are the intermediaries on the investment chain that directly face retail investors. Unlike the pooled fund front office, the degradation at this node **does not depend on a "black box"**—even when the underlying product is transparent and the mapping chain is intact, the node systematically provides biased advice due to commission incentives, the absence of suitability standards, and the void of fiduciary duty. This is the typical form of Pathway ②: the node possesses the capacity to exercise meta-audit capacity (A), but the incentive structure induces it to produce narrative load (ϕ).

Catering Rather Than Correcting Bias. Mullainathan, Noeth, and Schoar (2012) used trained "audit customers" to conduct an audit experiment, finding that advisors "fail to de-bias their clients and often reinforce biases that are in their interests," and "encourage returns-chasing behavior and push for actively managed funds that have higher fees, even if the client starts with a well-diversified, low-fee portfolio." The mechanism is not that advisors provide "bad advice," but that when client biases align with the advisor's interests, the advisor chooses to reinforce rather than correct them.

Commission-Driven and "Insufficient Disclosure." Anagol, Cole, and Sarkar (2017), in a series of field experiments in the Indian life insurance market, found that agents "overwhelmingly recommend unsuitable, strictly dominated products, which provide high commissions to the agent," and "cater to the beliefs of uninformed consumers, even when those beliefs are wrong." More critically, when regulation required the disclosure of commissions for a certain type of product, agents switched to recommending "alternative products with high commissions but no disclosure requirement"—indicating that disclosure alone is insufficient to eliminate incentive distortions; agents circumvent by maintaining high-commission structures through alternative products.

Not "A Few Bad Apples," but Market Structure. Egan, Matvos, and Seru (2019) constructed a database covering all financial advisors in the US from 2005–2015, finding that "Seven percent of advisers have misconduct records, and this share reaches more than 15% at some of the largest advisory firms"; about one-third of offenders are repeat offenders; misconduct is "concentrated at firms with retail customers and in counties with low education, elderly populations, and high incomes"; and offenders who are fired are often rehired by another group of firms with similarly high misconduct rates ("matching on misconduct"). This elevates the sales node from an isolated case to a **structural node with a labor market and organizational hierarchy**: certain institutions are, in a sense, "specializing in misconduct," targeting less sophisticated retail customer bases.

Institutional Design Changes Product Supply. Egan, Ge, and Tang (2022), in the variable annuities market, found that "variable annuity sales are roughly four times as sensitive to brokers' financial interests as to investors"'; after the 2016 proposed fiduciary rule from the Department of Labor, "sales of high-expense variable annuities fell by 52%," and sales became more fee-sensitive, with supply shifting towards lower-cost products. This pulls "the problem with the sales node" back from psychology/ethics to how institutional design can change product supply and sales structure.

Bhattacharya, Illanes, and Padi (2025, *Econometrica*), utilizing transaction-level data on deferred annuities and interstate differences in common law fiduciary duty, found that fiduciary duty "raises risk-adjusted returns by 25 bp and leads to a 16% decline in the entry of affected firms," and that the return improvement "arises from a

change in the set of transacted products"—fiduciary duty is not a moral declaration but substantively improves returns by raising the cost of low-quality advice and altering entry/exit and the product set.

These five pieces of evidence span three scenarios—US investment advisors, Indian life insurance, and US variable annuities—all pointing to the same mechanism, but **the respective boundaries must be preserved**: one cannot write from this that "all wealth management sales inevitably mislead clients," nor treat Chinese bank relationship managers, Indian life insurance agents, and US investment advisors as entirely isomorphic; Anagol et al.'s follow-up agent survey further shows that poor advice reflects both commission incentives and agents' limited product knowledge—the capacity conditions of nodes are not equally present in every setting, and this chapter's focus is on how incentive structures induce the abandonment of existing capacity. Mullainathan, Noeth, and Schoar (2012) is a US advisory audit experiment, Anagol, Cole, and Sarkar (2017) is Indian life insurance, and Egan, Ge, and Tang (2022) is US annuities—the scenarios are different. Structurally, however, they share Pathway ②: the node possesses meta-audit capacity (A), but the incentive structure of commissions/suitability/fiduciary duty induces it to produce narrative load (ϕ, accommodating rhetoric, recommending inferior products). This is a degradation mechanism distinct from the structural compulsion of pooled funds (Pathway ①).

§6.3 Current Cross-National Mechanism of Pathway ② (II): Asset Management Nodes

Asset management nodes (fund managers/investment managers) should be treated separately from sales nodes: the degradation of the former **does not primarily rely on "misleading clients"** but is induced by incentives such as AUM fees, flow-performance relationships, and career concerns, together with strategic behavioral tendencies such as window dressing towards $A \to \phi$. This is another sub-mechanism of Pathway ② and should be distinguished from commission-driven misleading at the sales end.

Arjaliès, Grant, Hardie, MacKenzie, and Svetlova (2017), in *Chains of Finance*, traced the most systematic conflicts of interest along the investment chain, citing the classic formulation by Chevalier and Ellison (1997): investors want asset management companies to "maximize risk-adjusted fund returns," but the income of the company and its executives "is usually determined much more directly by the total amounts of capital they manage and the fees they charge," thus they have "an incentive to take actions that increase the inflow of investments" (Chevalier & Ellison, 1997), even if these actions harm investor returns—for example, expanding fund size beyond the scale at which the strategy has genuine investment opportunities. This is precisely the node-level mechanism by which meta-audit capacity (A) is induced to be replaced by narrative load (ϕ—the discursive operation of substituting scale-and-performance narratives for risk explanation): assets-under-management (AUM) fee and ranking incentives induce scale expansion and portfolio dressing, and these behaviors substitute scale narratives for independent judgment about risk-adjusted returns.

Ma, Tang, and Wang (2024), in their review of mutual fund incentives, categorize these mechanisms into three types: implicit incentives (fund inflows, career concerns), explicit incentives (advisory contracts, portfolio manager compensation), and other incentives (managerial ownership, family affiliation, organization structures, regulations, and business ties); they also point out that portfolio disclosure, intended to increase transparency, may also induce strategic behaviors like window dressing and portfolio pumping—"more disclosure" does not equal "more audit capacity." As a review, it is suitable for providing a framework and navigating the literature, but not for independently supporting a strong empirical proposition about a specific behavior; its double-layered delegation structure of investors → fund advisors → portfolio managers should not be mechanically extrapolated

to all financial products.

Asset management nodes share the logic of Pathway ② with sales nodes (having capacity but being induced by incentives to abandon meta-audit capacity), but the sub-mechanisms differ: the sales end is driven by commissions and suitability; the asset management end by AUM and performance flows. Together, they form the complete picture of Pathway ② degradation along the investment chain.

§6.4 After Net-Value Transformation: The Current Predicament of Chinese Front-Office Practitioners (Pathway ②)

After the new asset management regulations broke implicit guarantees and implemented net-value-based management, the net values of wealth management products began to fluctuate. The "redemption wave" at the end of 2022 was a quintessential moment when fiber was forcibly made visible: adjustments in the bond market led to widespread "breaking the net" of bank wealth management products—net value falling below the 1 yuan par value. Investors accustomed to "principal protection and guaranteed returns" redeemed en masse, and redemptions further exacerbated bond sell-offs, forming a negative feedback loop.

For relationship managers, this was a sudden liquidation of cognitive debt (D_c). Previously, the pseudo-small fiber of "expected yield" had masked the true volatility of the underlying assets; now, net-valuation directly exposed that volatility to clients. What relationship managers faced was not an abstract "fiber inflation"—but concrete client complaints, performance assessment pressure, and a crisis of professional trust. The products he had once described as "very safe" lost money, and he needed to explain why to the client.

It is crucial to note: the front-office predicament after the net-value transformation differs from that of the pooled fund era. Pooled funds were Pathway ①'s "Structural Compulsion" (the mapping chain was technically severed, rendering tracing impossible); net-value transformation requires products to be measured at fair value, making volatility visible, and the underlying information is no longer institutionally obscured by the commingling mechanism of pooled funds. The difficulty for the current front office is no longer "structurally unable to see the underlying," but a variant of Pathway ②: under the incentives of sales targets and client retention, the front office assumes the new duty of "explaining volatility and absorbing friction"—what he needs to do is not to regain an audit capacity severed by structure, but rather to decide whether, in the new environment where net value is visible, he is willing to provide independent risk explanations for clients. This shift precisely illustrates that after the termination of the specific "black box" product structure of pooled funds, the predicament of the front office did not disappear but shifted from Pathway ① (Structural Compulsion) to Pathway ② (Incentive Induction).

Regarding temporality, the redemption wave at the end of 2022 was an acute moment when this Pathway ② predicament became manifest, not the permanent state of the front office; the norm for the current front office is closer to the "routine dissonance" distinguished in §6.5—low-intensity, persistent friction between net value volatility and client communication, rather than the concentrated liquidation of the 2022 type.

§6.5 Cognitive Dissonance and the Reconstruction of Meta-Audit Capacity (A)

The net-value transformation pushed relationship managers into a structural cognitive dissonance: a fundamental mismatch emerged between their professional skill (selling "safe" wealth management products) and the new institutional environment (products are no longer "safe," net values fluctuate with the market).

According to a People's Daily Online report in November 2025, the labor union of overseas Chinese-funded and foreign-funded banks in Pudong New Area, in the process of establishing an employee psychological service system, disclosed the results of an employee psychological status survey covering 753 employees across 11 institutions, showing that career development and compensation were the core sources of stress (these data come from a labor union disclosure, relayed by media, and are not the original text of an independently verifiable formal investigation report). The number of registered professionals in the securities industry saw a net decrease of 18,600 throughout 2024 and a further net decrease of nearly 8,000 in 2025 (Securities Association of China, 2026; the 2024 figure comes from the Association's China Securities Industry Development Report 2025 as relayed by China Securities Journal, and the 2025 figure is a journalist's tally, cross-verified against SAC data)—the contraction of industry employment and the bank employee stress survey together form the background picture of an industry under pressure, but neither can, on its own, identify the impact of the net-value reform or of cognitive debt liquidation, nor directly represent the situation of bank relationship managers. Evidence from a specific business line points the same way: according to the SAC's 2025 Statistical Report on the Securities Research Report Business, the total number of research personnel at securities firms fell by 3.68% year-on-year in 2025, institutional client trading commissions fell by 17.55%, and total industry public-fund commission allocations fell to half their 2021 level—the contraction of the research line tracks its revenue-side pressure from the public-fund fee reform, and likewise cannot separately identify the impact of the net-value reform. For firsthand narrative evidence from within the industry, see Response 1 (Derman, 2004) and Response 3 (Smith, 2012; Neely, 2022) below.

In the fiber framework, what relationship managers face is not a single event but cognitive pressure operating on two timescales: routine cognitive dissonance is the professional norm for relationship managers after the net-value transformation—the low-intensity, persistent friction between daily net value fluctuations and client inquiries; the micro-liquidation of cognitive debt occurs during concentrated outbreak moments triggered by market adjustments, such as the redemption wave at the end of 2022. Only the latter constitutes "liquidation"; the former is the everyday manifestation of debt—indiscriminately classifying routine dissonance as "liquidation" would over-extend the concept of liquidation and diminish its analytical sharpness.

In the fiber framework, relationship managers face three possible responses:

Response 1: Reconstruction of Meta-Audit Capacity (A). Some relationship managers begin to study underlying asset analysis, bond market knowledge, and portfolio management logic, rebuilding independent risk explanation and risk communication capabilities (A) within the new environment where net value is visible. While the long-term atrophy of this capacity indeed has historical causes from the pooled fund era, what is being reconstructed now targets the incentive pressures of Pathway ② post-net-value transformation, not the structural blindness of the old pooled fund structure. This is a positive effect of institutional calibration: when the institution no longer permits narrative load (ϕ) to substitute for meta-audit capacity (A), practitioners are forced to reinvest in the latter. The "Guidelines for Performance Assessment Management of Fund Management Companies" issued by the Asset Management Association of China (AMAC) in April 2026 institutionalize requirements such as deferred performance-based compensation (no less than three years), mandatory co-investment, and linking sales personnel assessments to investor gains and losses—these reforms provide an incentive foundation at the institutional level for a $\phi \to A$ reversal.[4] The Guidelines apply to fund management

[4]Chapter 18, §1.5, from the perspective of compensation incentives—corresponding to the role audit level (the incentive conditions for role audit capacity (A_{inst}, Chapter 3, §2.4))—diagnoses the first half of the same pressure transmission chain: how convex compensation compresses multi-dimensional risk into a one-dimensional scalar. This section unfolds along the second half of the

companies; the corresponding rules for the banking wealth management line are still evolving—they are cited here as an institutional signal of the incentive-reform direction on the asset management side.

Derman (2004), in *My Life as a Quant*, provides a cross-industry, firsthand account of meta-audit capacity (A) reconstruction: a theoretical physicist who, after five years of industrial research at Bell Labs, moved to Goldman Sachs in 1985 and later co-designed the Black-Derman-Toy interest rate model. The memoir repeatedly emphasizes two core lessons: first, financial models are never analogies to physical laws—"in physics you're playing against God … [i]n finance, you're playing against God's creatures"; second, the true responsibility of a quant is not to make a model give the "right answer," but to clearly report to users the model's assumptions and omissions (the essence of what Derman and Wilmott later crystallized in the 2009 *Financial Modelers' Manifesto* as the "Modelers' Hippocratic Oath"). This constitutes an industry insider's firsthand sample of meta-audit capacity (A) reconstruction at the individual level—reconstructing meta-audit capacity is not a return to some "original truth" but the establishment of a systematic vigilance towards the "inevitable failure of models." In the fiber framework, this corresponds to a practitioner reinvesting in independent audit capacity after realizing the systematic substitution of meta-audit capacity by narrative load.

Response 2: Transformation of Narrative Load (ϕ). Another segment of relationship managers adjusts their sales rhetoric from "principal protection and guaranteed returns" to "long-term holding," "dollar-cost averaging strategy," "asset allocation"—a new narrative load replaces the old one, but an independent audit of the true state remains fundamentally absent at the bottom. The narrative load has changed its garb; meta-audit capacity remains absent.

Response 3: Exit. When the cost of cognitive dissonance exceeds the occupational benefit, practitioners choose to leave. The continued contraction in the number of practitioners may be related to the pressure of cognitive debt liquidation, but reading it directly as a price signal of liquidation exceeds what the data can identify. Exit, as an individually rational choice, may have an aggregate effect at the macro scale: the continued exit of practitioners **may** reduce the supply of independent meta-audit capacity within the industry, depressing the average tenure and experience density of the practitioner cohort—but this aggregate effect is currently only suggested directionally by individual cases (Smith, Neely) and attrition data, and is not yet sufficient to be concluded as a definitively proven stable macro-consequence. Even so, its structural logic can be stated as follows: if practitioners under the incentive pressures of Pathway ② find that the cost of cognitive liquidation exceeds the benefit of remaining in the industry, exit is the rational choice—this is what the mechanism implies, not an assertion about exiters' actual motives.

The phenomenon of practitioner exit is not unique to the Chinese market. Two pieces of cross-national evidence point to pressure outcomes similar to those described in this subsection, but their evidentiary weight must be carefully delimited—what they share with the Chinese front office is a "structural homology" (after narrative load (ϕ) overwhelms meta-audit capacity (A), practitioners bear professional insecurity and exit pressure), not a mechanism precisely mapped to "mapping chain breakage → structurally compelled exit." The two pieces of evidence complement each other in grade: Smith's (2012) resignation letter is a **signal event**—it publicized a cognitive dissonance normally absorbed internally by institutions, but its significance lies not in statistical

same chain—starting from the scenario where institutional calibration forces fiber to become visible, it analyzes how practitioners are compelled, amidst cognitive dissonance, to rebuild a meta-audit capacity that has been rendered obsolete. The two chapters examine different facets of the same class of actors (financial practitioners) under institutional pressure: the compensation incentive perspective vs. the cognitive fiber perspective.

representativeness; Neely's (2022) **systematic ethnography**, based on in-depth interviews with dozens of practitioners, provides systematic qualitative evidence. Both corroborate the same structure from different angles.

Neely (2022), in *Hedged Out*, based on in-depth interviews with Wall Street hedge fund professionals, points out that Wall Street's "elite-periphery" talent structure is also driven by a structural filtering mechanism: the role drift of practitioners from "financial stewards" (creating long-term value for clients) to "Flash Boys" traders (capturing short-term profits for the institution) continuously separates individual skills from systemic risk exposure—when the industry cycle turns downward, peripheral practitioners are the first to be squeezed out, and **exit is, to a large extent, the result of structural pressures, not a purely individual choice**.

Smith (2012), in *Why I Left Goldman Sachs*, revisits the controversy triggered by his open resignation letter published in the *New York Times* in March 2012: as a 12-year veteran and vice president at Goldman Sachs, he describes the firm's cultural drift from "Our clients' interests always come first" to "actively trying to take advantage of clients"—he dates the turning point of this drift to the full substitution of long-term fiduciary duty by a "profit-at-all-costs model" after the 2008 crisis. This is precisely the concrete experience, at the individual level, of narrative load substituting for meta-audit capacity: when an institution internalizes narrative load to the cultural level, it becomes harder for individuals who retain meta-audit capacity to stay.

These two cross-national cases share structural homology with the net decrease of approximately 26,500 professionals in the Chinese securities industry over two years (as described in this section)—if this interpretation holds, the costs of institutional friction are borne by the practitioners at the terminal nodes of the mapping chain, whether that terminal is in New York or Beijing.

§6.6 The Complete Micro-Transmission Chain

The bank wealth management relationship manager of the pooled fund era (historical sample of Pathway ①) illustrates a complete transmission chain of fiber inflation from macro to micro. It must first be clarified: this chain is a historical illustration set against the "pooled fund structure" that has been institutionally terminated by the net-value transformation and cannot be directly applied to the current front office (whose mechanism is Pathway ②, see §6.4); for the transmission regarding sales and asset management nodes under Pathway ②, see §6.2 and §6.3, respectively:

1. **Mapping Chain Structural Breakage** (L0): Pooled funds sever the mapping chain between underlying assets and investor readings; the relationship manager is situated at the breakpoint.
2. **Incentive Structure Bias** (L1): Sales-oriented KPIs cause the relationship manager's governance clock to be captured by the sales cycle, systematically suppressing the motivation to audit.
3. **Narrative Load Replaces Meta-Audit Capacity** (L2): Institutional endorsement and faith in implicit guarantees become tools of pseudo-calibration—the relationship manager substitutes narrative for audit, and the investor substitutes trust for verification.
4. **The Shock of Institutional Calibration**: The new asset management regulations break implicit guarantees and mandate net-value management, exposing all at once the fiber inflation that had long accumulated in the hidden subspace.
5. **Micro-Liquidation of Cognitive Debt**: The relationship manager endures client complaints, professional crisis, and cognitive dissonance—this is the micro-cost of macro fiber inflation.

The causal logic of this transmission chain is entirely internal to the fiber framework: it is not that "an industry

downturn causes difficulties for practitioners," but that "mapping chain structural breakage → institutional calibration forces fiber to become visible → practitioners bear the brunt of the micro-costs of cognitive debt liquidation." However, this chain, set against the background of pooled funds, is a historical form of Pathway ①; after the net-value transformation, the same front-office practitioner now faces the incentive structure of Pathway ② (§6.4), sharing the same degradation logic as sales and asset management nodes (§6.2, §6.3). The common conclusion of both pathways is: the practitioners' predicament is not a moral failure but a structural consequence marked by the MCAF as a diagnostic chart of accounts—per Chapter 3's positioning, the meta-formula does not solve, but the defensive-side failure conditions it marks here become visible as practitioners' concrete circumstances.

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§7 Conclusion: The Micro-Costs of Fiber Inflation

This chapter descends the macro diagnoses of Chapters 17 through 20 to the micro level, revealing two layers of micro-consequences of fiber inflation.

Product Level: Three financial products present three different fiber states—the fiber of pooled fund products is artificially compressed by fund commingling and implicit-guarantee expectations (pseudo-small fiber), the fiber of cryptocurrencies is the total set (no underlying asset to anchor to), and the fiber of derivatives inflates with the number of nested layers, yet the underlying true state (T) exists. Three calibration models—penetration + prohibition + net-value-based management (New Asset Management Regulations), prohibiting noise from entering (cryptocurrencies), and prudent development + chain length limitation (derivatives)—correspond to three different epistemological stances. Calibration requires continuous work; once interrupted, the system slides back along the entropic direction toward a de-anchored state.

Individual Level: This chapter does not presuppose "several categories of practitioners" but identifies, along the investment/mapping chain, the key intermediary nodes where the $A \to \phi$ degradation actually occurs, distinguishing two degradation pathways. **Pathway ① (Structural Compulsion)**: Institutional design renders practitioners **unable** to exercise meta-audit capacity (A). Bank wealth management relationship managers in the pooled fund era serve as its historical sample—but this is a product of a historically specific period prior to the 2018 net-value transformation under the new asset management regulations, has been institutionally terminated, and cannot be used to bear the burden of arguing for "the structural predicament of current practitioners."

Pathway ② (Incentive Induction): Nodes **possess the capacity but are induced by incentive structures to abandon** meta-audit capacity (A); its empirical demonstration is found in Chapter 18's analysis of convex compensation and ranking tournaments. Pathway ② remains cross-nationally prevalent in the present: sales/distribution nodes cater to rather than correct client biases due to commissions, suitability deficiencies, and the void of fiduciary duty (Mullainathan, Noeth & Schoar, 2012; Anagol, Cole & Sarkar, 2017; Egan, Matvos & Seru, 2019; Egan, Ge & Tang, 2022; Bhattacharya, Illanes & Padi, 2025—spanning US advisory, Indian insurance, and US annuity scenarios); asset management nodes, driven by AUM, flow-performance, career concerns, and window dressing, swell their scale and dress up portfolios (*Chains of Finance* citing Chevalier & Ellison, 1997; Ma, Tang & Wang, 2024 review).

The two pathways correspond to different calibration strategies: Pathway ① requires reforming institutional design (breaking implicit guarantees, as with the New Asset Management Regulations); Pathway ② requires

reforming incentive structures (compensation deferral and clawback mechanisms, as well as fiduciary duty, as with the 2026 AMAC "Guidelines for Performance Assessment Management"; the US fiduciary duty rule provides a sample of an institutional attempt amidst ongoing controversy, rather than a mature path of linear implementation). After the net-value transformation, the Chinese front office has shifted from "selling expected returns" to the new duty of "explaining net value volatility and absorbing client friction"—this is the current form of Pathway ② in the Chinese context (rather than a continuation of the pooled fund Pathway ①), and the front office bears the brunt of the micro-costs of cognitive debt liquidation.

The pseudo-small fiber of products is borne by investors; the cognitive dissonance of practitioners is borne by the practitioners themselves. The common feature of both is: **the bearer of the cost is not the maker of the cost**. Structure manufactures fiber inflation; the individual bears its consequences.

The next chapter (Chapter 22) will switch from the fiber framework (static structural diagnosis) to the MCAF framework (dynamic debt tracking), extracting MCAF's structural diagnoses from four financial history cases—the critical phase transition of the manifestation function $M(\tau)$, the impossibility of purely defensive-side intervention, and the feasibility of structural intervention combined with a gradualist governance clock.

Glossary of Term Correspondences

This chapter descends the macro diagnoses established in Chapters 17 through 20 to the micro level of financial products and practitioners. The correspondences between financial phenomena and the core concepts of the fiber framework and MCAF framework are as follows:

Object of Diagnosis in Chapter 21	Fiber Framework Correspondence	MCAF Framework Correspondence
Pooled Funds—"Expected annualized yield 4.5%"	Pseudo-Small Fiber (L2)—precise reading masks the risk uncertainty of the underlying assets	D_c^{shadow}—the gap between true risk and observable risk
New Asset Management Regulations—Breaking Implicit Guarantees	Forcing $D_c^{obs} \to D_c^{true}$	Policy-driven manifestation of D_c^{shadow}
New Asset Management Regulations—Look-Through Supervision	L0 Mapping Chain Truncation—limiting the number of layers the mapping chain can stack	Increase in the effective value of A
New Asset Management Regulations—Prohibiting Pooled Funds	Severing the self-referential structure of $\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$	Making the Λ channel traceable
New Asset Management Regulations—Net-Value-Based Management	Pseudo-small-fiber exposed amidst price fluctuations	Net-value fluctuations provide a signal for testing pre-existing narratives; their interpretation remains subject to valuation methods and incentives

Object of Diagnosis in Chapter 21	Fiber Framework Correspondence	MCAF Framework Correspondence
Uncollateralized Cryptocurrencies (Bitcoin)	Fiber is the Total Set—$\pi^{-1}(r)$ is compatible with all possibilities because T does not exist	—
Fiat-Collateralized Stablecoins (USDT/USDC)	Mapping function exists, anchor is traceable—stablecoin price → USD reserves → Federal Reserve balance sheet → US government's taxing power	The ultimate anchor of $\|L\|$ is sovereign power
Derivatives—Restriction on the Number of Nested Layers	L0 Mapping Chain Length Limitation—the clock must be traceable back to the sun within a finite number of steps	The structural complexity of C is institutionally constrained
Relationship Manager's "Bank Endorsement"	ϕ replaces A—narrative load replaces meta-audit capacity	Where the front office lacks verification conditions or substitutes unverified institutional endorsement for risk explanation, the exercise of A is suppressed
Relationship Manager's Cognitive Dissonance	Micro-cost of L2 Pseudo-Calibration—after ϕ replaces A, institutional calibration forces fiber to become visible	Micro-liquidation of D_c^{shadow}
Sales/Distribution Nodes—Commission-driven "catering rather than correcting bias"	ϕ replaces A (Pathway ② Incentive Induction)—node possesses audit capacity, but incentives induce reinforcement rather than correction of client biases	A systematically suppressed by commission/suitability incentive structures (distinct from the institutional compulsion of Pathway ①)
Asset Management Nodes—Scale-driven impulse from AUM/performance flows	ϕ replaces A (Pathway ② Incentive Induction)—swelling scale and portfolio dressing replace independent audit of risk-adjusted returns	A systematically suppressed by AUM/ranking incentive structures

References

Ahmed, R. & Aldasoro, I. (2025). Stablecoins and Safe Asset Prices. *BIS Working Papers*, No. 1270. https://www.bis.org/publ/work1270.htm

Anagol, S., Cole, S. & Sarkar, S. (2017). Understanding the Advice of Commissions-Motivated Agents: Evidence from the Indian Life Insurance Market. *The Review of Economics and Statistics*, 99(1). https://doi.org/

10.1162/rest_a_00625

Arjaliès, D.-L., Grant, P., Hardie, I., MacKenzie, D. & Svetlova, E. (2017). *Chains of Finance: How Investment Management is Shaped*. Oxford: Oxford University Press.

Asset Management Association of China. (2026-04-17). *Guidelines for Performance Assessment Management of Fund Management Companies* (Zhongjixiefa [2026] No. 9). (In Chinese.) https://www.amac.org.cn/xwfb/xhyw/202604/t20260417_27594.html

Auer, R., Lewrick, U. & Paulick, J. (2025). DeFiying Gravity? An Empirical Analysis of Cross-Border Bitcoin, Ether and Stablecoin Flows. *BIS Working Papers*, No. 1265. https://www.bis.org/publ/work1265.htm

Bhattacharya, V., Illanes, G. & Padi, M. (2025). Fiduciary Duty and the Market for Financial Advice. *Econometrica*, 93(4). https://doi.org/10.3982/ECTA18492

Bloomberg News. (2025a). Blackstone, Apollo and Others Sell Bonds Backed by Private Credit at Fastest Pace Ever. *Financial Advisor Magazine*, 2025-09-24. https://www.fa-mag.com/news/blackstone--apollo-and-others-sell-bonds-backed-by-private-credit-at-fastest-pace-ever-84197.html

Bloomberg News. (2025b). US CLO Sales Reach Fresh Record in 2025 as Demand Stays Strong. *Bloomberg*, 2025-12-17. https://www.bloomberg.com/news/articles/2025-12-17/us-clo-sales-reach-fresh-record-in-2025-as-demand-stays-strong (open relay: https://alphatest10.wordpress.com/2025/12/17/bbg-us-clo-sales-reach-fresh-record-in-2025-as-demand-stays-strong/)

Bookstaber, R. (2007). *A Demon of Our Own Design: Markets, Hedge Funds, and the Perils of Financial Innovation*. Hoboken, NJ: John Wiley & Sons.

Bu, Z. X. (2021). Study on the Reasons and Countermeasures for the Extension of the New Asset Management Regulations. *Financial Theory & Practice*, 2021(4). (In Chinese.)

Cai, G. W., Zhu, Y. D. & Zhou, T. (2022). Has the Wealth Management Product Market Served the Real Economy?—A Quasi-Natural Experiment Based on Investment Income Tax Reform Policies. *Statistical Research*, 39(6). (In Chinese.)

Central Committee of the Communist Party of China. (2025, October). *Recommendations for Formulating the Fifteenth Five-Year Plan for National Economic and Social Development*. (In Chinese.) https://www.mofcom.gov.cn/syxwfb/art/2025/art_d5da513be3fe491582dd17eae6d805d9.html

Chen, G. J., Jiang, X. Y., Liu, Y. Z. & Zhao, X. Q. (2021). Asset Opacity, Regulatory Arbitrage, and Bank Systemic Risk. *Journal of Financial Research*, 2021(3). (In Chinese.)

Chevalier, J. & Ellison, G. (1997). Risk Taking by Mutual Funds as a Response to Incentives. *Journal of Political Economy*, 105(6). https://doi.org/10.1086/516389

China Financial Information Network. (2026). 2025 Asset Securitization Development Report. *China Financial Information Network*, 2026-04-22. (In Chinese.) https://www.cnfin.com/zs-lb/detail/20260422/4402765_1.html

Derman, E. (2004). *My Life as a Quant: Reflections on Physics and Finance*. Hoboken, NJ: John Wiley & Sons.

Derman, E. (2011). *Models.Behaving.Badly: Why Confusing Illusion with Reality Can Lead to Disaster, on Wall Street and in Life*. New York: Free Press (US) / Chichester: John Wiley & Sons (UK).

Derman, E. & Wilmott, P. (2009). *The Financial Modelers' Manifesto*. wilmott.com, 2009-01-07. https://wilmott.com/financial-modelers-manifesto/

Egan, M., Ge, S. & Tang, J. (2022). Conflicting Interests and the Effect of Fiduciary Duty: Evidence from Variable Annuities. *The Review of Financial Studies*, 35(12). https://doi.org/10.1093/rfs/hhac047

Egan, M., Matvos, G. & Seru, A. (2019). The Market for Financial Adviser Misconduct. *Journal of Political Economy*, 127(1). https://doi.org/10.1086/700735

Fang, Y. & Chen, M. (2019). Economic Fluctuations, Bank Risk-Taking, and China's Financial Cycle. *The Journal of World Economy*, 42(2). (In Chinese.)

Gao, R., Chen, C., Zeng, H. & Gong, L. T. (2018). Credit Constraints, Shadow Banking, and Monetary Policy Transmission. *Economic Research Journal*, 2018(12). (In Chinese.)

Guotai Haitong Asset Management. (2026). 2025 REITs Market Annual Review: Steady Progress in Normalized Issuance and Complementary Development of a Multi-Level Market. (In Chinese.) 2026-03-25. https://www.10100.com/article/125275256

Ji, M. & Li, H. J. (2018). The Impact of Off-Balance Sheet Wealth Management and Asset Management Businesses of Banks on the Effectiveness of China's Monetary Quantity Control. *Finance and Economy*, 2018(7). (In Chinese.)

Kim, S. I. (2024). Unveiling Mispricing Risks: Nonlarge Homogeneous Portfolio Factor Copula Models for Enhanced Valuation of Subordinated Loan Securitization. *Journal of Futures Markets*, 44(10). https://doi.org/10.1002/fut.22535

Li, J. S. (2025). Shadow Banking, New Asset Management Regulations, and Bank Risk-Taking. *National Circulation Economy*, 2025(2). (In Chinese.)

Liu, C., Pang, Y. C. & Liu, L. Y. (2022). Structural Monetary Policy, Financial Regulation, and the Efficiency of Interest Rate Transmission—Evidence from China's Bond Market. *Economic Research Journal*, 2022(1). (In Chinese.)

Lu, C., Zhao, T. Y. & Niu, Y. H. (2025). The Unintended Cost of Financial Regulation in China: The Impact of the New Regulation on Asset Management on Private Firms' Credit Spreads. *Pacific-Basin Finance Journal*, 93. https://doi.org/10.1016/j.pacfin.2025.102860

Ma, L., Tang, Y. & Wang, Y. (2024). Managerial Incentives in the Mutual Fund Industry. *Oxford Research Encyclopedia of Economics and Finance*. https://doi.org/10.1093/acrefore/9780190625979.013.889

MacKenzie, D. (2006). *An Engine, Not a Camera: How Financial Models Shape Markets*. Cambridge, MA: MIT Press.

Miller, M. H. (1995). Do We Really Need More Regulation of Financial Derivatives? *Pacific-Basin Finance Journal*, 3(2–3). https://doi.org/10.1016/0927-538X(95)00008-9

Minsky, H. P. (1982/2016). *Can "It" Happen Again? Essays on Instability and Finance*. With a new foreword by Jan Toporowski. London & New York: Routledge Classics.

Minsky, H. P. (1986). *Stabilizing an Unstable Economy*. New Haven: Yale University Press.

Mullainathan, S., Noeth, M. & Schoar, A. (2012). *The Market for Financial Advice: An Audit Study*. NBER Working Paper No. 17929. https://doi.org/10.3386/w17929 [Preprint, not yet peer-reviewed]

National Association of Financial Market Institutional Investors. (2026, April). *China Bond Market Reform and Development Report (2026)*. (In Chinese.) https://www.nafmii.org.cn/xhdt/202604/t20260414_325341.html

National Financial Regulatory Administration. (2025, October). *Administrative Measures for Asset Management Trusts (Draft for Comments)*. (In Chinese.) https://www.nfra.gov.cn/cn/view/pages/ItemDetail.html?docId=1231597&itemId=951&generaltype=2

National Financial Regulatory Administration. (2025, December). *Administrative Measures for Information Disclosure of Asset Management Products of Banking and Insurance Institutions* (NFRA Order No. 10 of 2025). (In Chinese.) https://www.gov.cn/zhengce/202512/content_7053030.htm

Neely, M. T. (2022). *Hedged Out: Inequality and Insecurity on Wall Street*. Oakland: University of California Press.

Pan, G. S. (2025). Firmly Practicing Global Governance Initiatives, Continuously Advancing the Reform and Improvement of Global Financial Governance. *Qiushi*, 2025(18). (In Chinese.) https://www.qstheory.cn/20250914/a0a073b6f3d64ceab1f85e9c353d58ca/c.html

People's Bank of China, China Banking and Insurance Regulatory Commission, China Securities Regulatory Commission & State Administration of Foreign Exchange. (2018, April). *Guiding Opinions on Regulating the Asset Management Business of Financial Institutions* (Yinfa [2018] No. 106). (In Chinese.) https://www.gov.cn/zhengce/zhengceku/2018-12/31/content_5433072.htm

People's Bank of China and seven other departments. (2026, February). *Notice on Further Preventing and Addressing Risks Related to Virtual Currencies and Related Matters* (Yinfa [2026] No. 42). (In Chinese.) https://www.csrc.gov.cn/csrc/c100028/c7614318/content.shtml

People's Daily Online. (2025, November 17). Labor Unions of Overseas Chinese-Funded and Foreign-Funded Banks in Pudong Take the Lead in Establishing Employee Psychological Service Systems. (In Chinese.) https://sh.people.com.cn/n2/2025/1117/c134768-41413685.html

SEC. (2025). Crypto Task Force Priorities. https://www.sec.gov/featured-topics/crypto-task-force

Securities Association of China. (2026). *China Securities Industry Development Report 2025*. Industry report (2024 data; the 2025 figures are a journalist's tally in the China Securities Journal report of 2026-02-19, cross-verified against SAC data). (In Chinese.) https://finance.eastmoney.com/a/202602193651497745.html

Securities Association of China. (2026, October). *2025 Statistical Report on the Securities Research Report Business*. Industry report (relayed by China Securities Journal, 2026-10-17, and China Business Journal, 2026-10-15). (In Chinese.) https://epaper.stcn.com/con/202610/16/content_2984153.html

Shao, X. J., Dong, D. D. & Hong, J. J. (2023). Research on the Pricing Mechanism of Chinese-Style Shadow Banking for Local Government Financing Platform Debt. *Journal of Financial Research*, 2023(4). (In Chinese.)

Si, D. K., Li, Y. J. & Li, X. L. (2022). Banking Competition in China and the Shadow Bankingization of Non-Financial Enterprises. *Journal of Financial Research*, 2022(8). (In Chinese.)

Smith, G. (2012). *Why I Left Goldman Sachs: A Wall Street Story*. New York: Grand Central Publishing (Hachette Book Group).

Soros, G. (1987/2003). *The Alchemy of Finance* (New edition with a new Introduction). Hoboken, NJ: John Wiley & Sons.

Standing Committee of the National People's Congress. (2026, August). *Law of the People's Republic of China on the Regulation of the Banking Industry (Draft Revision, Second Deliberation Draft)*. (In Chinese.) http://www.npc.gov.cn/c2/c30834/202608/t20260828_457167.html

Tang, S. & Rao, P. G. (2025). Strong Financial Regulation and Trade Credit Reallocation. *Journal of Quantitative & Technological Economics*, 2025(6). (In Chinese.)

U.S. Congress. (2025). GENIUS Stablecoins Act (S.1582). https://www.congress.gov/bill/119th-congress/senate-bill/1582/text

Wei, L., Wang, Y. T. & Zhang, W. (2018). Study on Cash Pool Operations and the Evolution of Shadow Banking. *Financial Development Review*, 2018(9). (In Chinese.)

Wen, X. X. & Su, N. F. (2018). Big Asset Management, Shadow Banking, and Monetary Policy Transmission. *Journal of Financial Research*, 2018(10). (In Chinese.)

Xiang, H. J. & Yan, Y. (2017). Research on the Development of Wealth Management Products, Interest Rate Liberalization, and Bank Risk-Taking. *Journal of Financial Research*, 2017(10). (In Chinese.)

Xiang, H. J. & Zhou, X. (2022). Shadow Banking and Its Regulation from the Perspective of Liquidity Hoarding. *Economic Research Journal*, 2022(3). (In Chinese.)

Xu, Z. (2018). Discussion on an Effective Financial Regulatory System. *Tsinghua Financial Review*, August 2018. (In Chinese.)

Zhang, J. M. (2022). Monetary Policy, Heterogeneity of Bank Risk-Taking, and Shadow Banking. *Economic Research Journal*, 2022(5). (In Chinese.)

Zhang, Y. L., Liang, W. & Yu, Y. J. (2023). The Formation and Governance Mechanism of Rigid Redemption in the Development of China's Shadow Banking. *Economic Research Journal*, 2023(2). (In Chinese.)

Zhao, D. P. & Wang, X. F. (2025). Strong Regulation of Shadow Banking and Bank Risk-Taking. *Studies of International Finance*, 2025(5). (In Chinese.)

Zhong, Z. (2018). Financial Regulatory Coordination: Proposal and Application of the Channel Theory. *Finance & Trade Economics*, 39(9). (In Chinese.)

Zhou, S. Y. & Wang, S. (2021). The Causes, Structure, and Systemic Risk of China's Shadow Banking. *Economic Research Journal*, 2021(7). (In Chinese.)